IS2212 Cloud Infrastructure and Applications

Assignment 02

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Scenario 1: E-Commerce Flash Sale Platform

1. Recommend a cloud deployment model (multi region? multi-AZ?) and justify your choice.

A Public Cloud, Multi-Region deployment on Amazon Web Services is recommended.

- Multi-Region ensures low latency for global users and keeps the platform alive even if an entire region goes down.
- Within each region, a Multi-Availability Zone (Multi-AZ) setup provides strong fault tolerance and high availability.
- This model gives rapid elasticity and pay-per-use benefits—perfect for extreme, short-lived flash-sale traffic spikes.

2. Identify micro-services to extract from the monolith.

- Product Catalog & Inventory – Handles massive read traffic during browsing.
- Static Content Delivery – Images and static assets served via a CDN to offload app servers.
- Shopping Cart – Isolated to scale independently and manage session state.
- Order & Payment Processing – Secure, critical service processed asynchronously for reliability and compliance.
- User Authentication & Session Management – Stateless service to allow fast horizontal scaling.

3. Draw or describe a high level cloud architecture supporting flash sale scale.

- **Networking:** VPC with public and private subnets.
- **Traffic Flow:**
  - Static assets served through a **CDN**.
  - Dynamic traffic goes through an Application Load Balancer (ALB) across multiple AZs.
- **Compute Layer:**
  - Micro-services run in Auto Scaling Groups (ASG) within private subnets.
- **Data & Decoupling:**
  - Managed databases with Multi-AZ and read replicas.
  - Heavy write operations (orders, payments) buffered using SQS or Kafka to protect databases and enable async processing.

4. Define auto scaling rules for identified through micro-services (include metrics & thresholds).

- Predictive Scaling – Use machine learning to forecast traffic bursts based on historical flash sale patterns and provision resources 15 – 30 minutes beforehand.
- Step Scaling – Configure a policy to add multiple instances rapidly when CPU Utilization exceeds 70% or Request Count spikes suddenly.

- Target Tracking – Maintain an average CPU Utilization of 50% to ensure enough headroom for unpredictable bursts.

5. Which caching strategies would you apply and where?

- Edge Caching (CDN) – Static assets cached globally to reduce latency and origin load.
- In-Memory Cache (Redis) – Product data, inventory, and sessions.
- Browser Caching – Cache headers for static content to reduce repeat requests.

6. What disaster recovery (DR) strategy fits this workload and why?

Warm Standby

- A Warm Standby in a second region keeps a scaled-down environment ready, enabling recovery in minutes to hours at moderate cost.
- For high-revenue flash sales, this balance minimizes downtime risk without full hot-site costs.
- Critical services can optionally run in Hot mode during major sale events.

Scenario 2: Online Learning System (Live Courses)

1. Propose an architecture that supports live streaming and video on demand (VOD).

A decoupled, microservices-based architecture on a Public Cloud (Amazon Web Services) deployed across Multiple Availability Zones (Multi-AZ) is recommended.

- Ingestion – Use a managed media ingestion service to receive live streams and transcode them into multiple formats for different devices.
- Delivery – Use Amazon CloudFront (CDN) to distribute both live streams and VOD globally with low latency.
- Compute:
  - Chat, quizzes, exams, and APIs run on EC2 instances in Auto Scaling Groups (ASGs) behind an Application Load Balancer.
- Decoupling – Compute and storage are fully separated. Video files are never stored on application servers. For that, we can use services like Amazon S3 or Azure Blob.

2. Explain how your proposal improves both live and recorded video delivery.

- Reduced Latency – CDN edge locations deliver content closer to learners, improving video start time and reducing buffering.
- Elasticity – The platform can scale from ~20 learners to 10,000+ concurrent users during live lectures or exams, then scale back down to optimize costs.
- Stability – CDN offloads traffic so origin servers aren't overwhelmed during peak events.

3. Recommend scaling strategies for live chat, transcoding, and API requests during peak times.

The system experiences periodic workloads (scheduled classes) and unpredictable spikes (exams, deadlines).

- Live Chat – Use Amazon SQS to decouple chat messages from backend processing, preventing API crashes during message bursts.
- API Scaling – Combine Scheduled Scaling (pre-warm before classes/exams) with Target Tracking based on CPU.
- Transcoding:
    - Handle asynchronously using worker instances triggered via SQS events.
    - Use Spot Instances in ASGs for cost-efficient transcoding since jobs are non-interactive.

4. Propose a storage architecture for live recordings, archives, and metadata (cost tiers).

- Live & Recent Recordings – Store in Amazon S3 Standard for fast, frequent access.
- Archived VOD – Move to S3 Standard-IA after 30 days to reduce cost while keeping low-latency access.
- Long-Term Archives & Compliance – Store in S3 Glacier Deep Archive for lowest-cost, long-term retention.
- Metadata – Store video metadata, user profiles, and chat logs in a managed database (RDS or DynamoDB).

5. How would you ensure high availability and continuity during exams?

- Multi-AZ Deployment – Ensures the system stays online even if one data center fails.
- Scheduled Scaling – Pre-warm infrastructure before known exam start times to avoid cold starts.
- Health Checks – ASGs automatically replace unhealthy instances.
- Disaster Recovery – A Warm or Hot DR site in a secondary region allows rapid failover during regional outages, which is critical during exams.

Scenario 3: Banking Application Modernization (Hybrid Cloud)

1. Explain why a hybrid cloud model is required.

A Hybrid Cloud model combining on-premises infrastructure with a Public Cloud (Amazon Web Services) is required.

- Regulatory and compliance requirements (e.g., CBSL, GDPR, PCI DSS, data residency laws) mandate that core banking systems and sensitive ledger data remain on-premises.
- A hybrid approach allows the bank to retain full control over regulated systems while using the public cloud's elasticity and scalability for modern, customer-facing services.
- This balances security, compliance, performance, and innovation without risking legacy systems.

2. Identify customer facing services suitable for cloud migration and their dependencies

Services to migrate to the cloud:

- Customer Dashboards & Analytics – Compute-intensive and benefit from elastic scaling.
- Notifications & Fraud Alerts – Event-driven micro-services capable of real-time processing.

- Customer APIs – For mobile and web banking access.

Dependencies:

- These services require secure, low-latency connectivity to the on-prem core banking system for real-time balance checks and transaction history.
- To prevent cloud traffic spikes from overwhelming legacy systems, requests should be decoupled using event-driven patterns with message queues (e.g., Amazon SQS or Kafka).

3. List the security functions and features you consider when giving a solution.

- Identity & Access Management (IAM) – Use IAM Roles to grant temporary, least-privilege access to cloud resources without hardcoded credentials.
- Encryption – Encrypt data at rest (EBS, S3) and in transit using TLS.
- Network Isolation – Deploy services in a VPC with:
  - Public Subnets for frontend services
  - Private Subnets for backend micro-services and databases
- Firewalls & Protection:
  - Use Security Groups (instance-level) and NACLs (subnet-level)
  - Add a Web Application Firewall (WAF) to protect against DDoS and web attacks

4. Describe secure connectivity between cloud micro-services and on-prem core banking.

Because the cloud micro-services must communicate with the on-premises core banking system, standard internet routing is not secure enough.

Recommended options:

- Site-to-Site IPSec VPN – You can create a secure, encrypted tunnel over the internet connecting the bank's Customer Gateway to the cloud VPC.
- AWS Direct Connect – For a bank requiring highly reliable, low-latency, and dedicated physical connectivity that bypasses the public internet entirely, Direct Connect is the ideal enterprise solution.
- If the bank scales to use multiple cloud regions or multiple VPCs in the future, an AWS Transit Gateway should be used as a central hub router to manage the connections between the VPCs and the on-premises datacenter.

5. Recommend a DR topology and justify RPO/RTO values.

Recommended DR Strategy - A Hot DR site or a highly robust Multi-AZ Architecture combined with a Warm DR site in another region.

Justification:

- RPO (Recovery Point Objective):
  - Must be near-zero (seconds to minutes) since losing transaction data is unacceptable.
- RTO (Recovery Time Objective):
  - Must be very short (minutes to hours) to maintain customer trust and service continuity.

This approach ensures financial accuracy, regulatory compliance, and high customer confidence, while controlling long-term operational costs.